Task-19:- Requirement Analysis

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for setting up a **multi-provider email alerting system integrated with Grafana**.

This document is intended for:

- DevOps Engineers
- System Administrators
- Project Evaluators

It provides a clear understanding of system functionality, integrations, and expected outputs.

1.2 Scope

In Scope:

- Setup of self-hosted mail server (Postfix on EC2)
- Configuration of Grafana SMTP settings
- AWS SES email configuration
- AWS WorkMail setup
- SNS integration for alert delivery
- Testing alert notifications

Out of Scope:

- Production-level scaling
- Advanced email security configurations (SPF, DKIM, DMARC)
- Multi-region failover setup

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SMTP	Simple Mail Transfer Protocol
SES	Simple Email Service

SNS	Simple Notification Service
EC2	Elastic Compute Cloud
MTA	Mail Transfer Agent
Grafana	Monitoring and alerting tool
Postfix	Mail server software
WorkMail	AWS managed email service

1.4 References

- AWS SES Documentation
- AWS SNS Documentation
- AWS WorkMail Documentation
- Grafana Official Documentation
- Postfix Documentation

1.5 Overview

This document includes:

- Project objectives and background
- Functional and non-functional requirements
- Use cases and workflows
- Risks, assumptions, and dependencies

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2. Project Overview

2.1 Project Objectives

- Set up a self-hosted SMTP mail server
- Configure Grafana to send alerts via SMTP
- Integrate AWS SES for email delivery
- Configure AWS WorkMail for enterprise email
- Implement SNS-based alert delivery
- Validate alert delivery across all providers

2.2 Business Goals

- Ensure reliable alert delivery system
- Evaluate multiple email providers
- Improve monitoring and incident response
- Reduce dependency on a single email provider
- Build scalable alerting infrastructure

2.3 Background

ABC Technologies uses Grafana for monitoring infrastructure. Currently, alert notifications are limited and not diversified.

To improve reliability and flexibility, the company wants to test multiple email delivery mechanisms including self-hosted SMTP, AWS SES, and AWS WorkMail.

2.4 Constraints

- AWS SES sandbox limitations
- Time constraint (4 hours completion)
- Limited budget (use free-tier services)
- SMTP ports may be restricted
- Domain/email verification required

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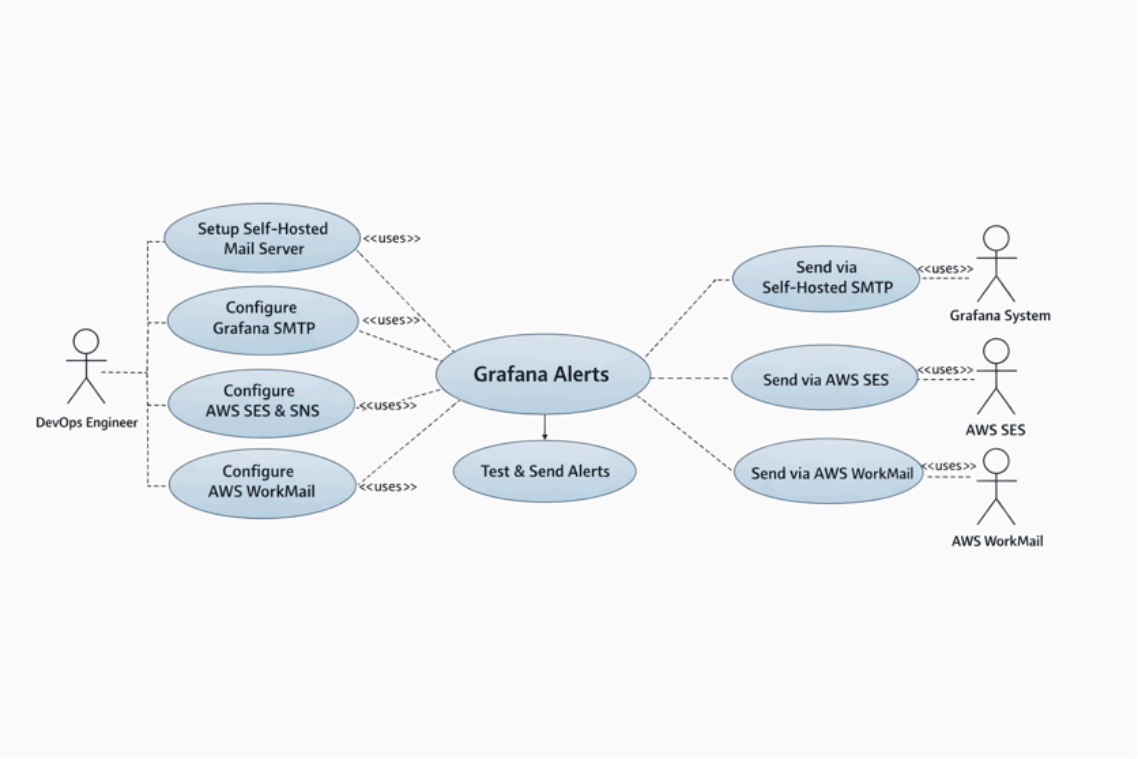
3. Functional Requirements

3.1 Use Case Diagram



Actors:

- DevOps Engineer
- Grafana System
- AWS Services

3.2 Use Cases

3.2.1 Use Case 1: Setup Self-Hosted Mail Server

Description:
Deploy and configure a mail server on EC2 using Postfix.

Actors:
DevOps Engineer

Preconditions:

- EC2 instance is running
- SSH access available

Postconditions:

- Mail server is configured and able to send emails

Flow of Events:

1. Launch EC2 instance
2. Install Postfix
3. Configure SMTP settings
4. Enable authentication
5. Test email sending

Alternative Flows:

- SMTP port blocked → Use alternate port
- Configuration error → Reconfigure Postfix

3.2.2 Use Case 2: Configure Grafana SMTP

Description:
Configure Grafana to use SMTP server for sending alerts.

Actors:
DevOps Engineer, Grafana

Preconditions:

- Grafana installed
- SMTP server running

Postconditions:

- Grafana sends alert emails

Flow of Events:

1. Open Grafana config file
2. Add SMTP details
3. Restart Grafana
4. Create test alert
5. Verify email delivery

Alternative Flows:

- Incorrect credentials → Fix configuration

- Email not received → Check logs

3.2.3 Use Case 3: Configure AWS SES

Description:
Set up AWS SES for sending emails.

Actors:
DevOps Engineer, AWS SES

Preconditions:

- AWS account available

Postconditions:

- SES can send verified emails

Flow of Events:

1. Verify email/domain
2. Configure sending permissions
3. Create SNS topic
4. Subscribe email
5. Send test message

Alternative Flows:

- SES sandbox restriction → Verify additional emails

3.2.4 Use Case 4: Configure AWS WorkMail

Description:
Set up WorkMail for receiving alert emails.

Actors:
DevOps Engineer, AWS WorkMail

Preconditions:

- Domain configured

Postconditions:

- WorkMail mailbox receives emails

Flow of Events:

1. Create organization
2. Create user mailbox
3. Configure email access
4. Subscribe to SNS
5. Test email delivery

Alternative Flows:

- Domain issue → Reconfigure domain
- Email not received → Check SNS subscription

3.2.5 Use Case 5: Alert Testing

Description:
Test alert delivery from Grafana.

Actors:
Grafana, DevOps Engineer

Preconditions:

- All integrations completed

Postconditions:

- Alerts received via all providers

Flow of Events:

1. Create alert rule (CPU, server down)
2. Trigger alert
3. Verify email delivery via:
   - SMTP
   - SES
   - WorkMail

Alternative Flows:

- Alert not triggered → Check rule
- Email failure → Debug provider

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4. Non-Functional Requirements

4.1 Performance Requirements

- Alerts should be delivered within a few seconds
- System should handle multiple alerts simultaneously

4.2 Security Requirements

- SMTP authentication required
- AWS IAM roles and permissions must be configured
- Secure access to EC2 instance

4.3 Usability Requirements

- Easy configuration and setup
- Clear logs for debugging
- Simple alert creation in Grafana

4.4 Reliability Requirements

- High success rate of email delivery
- Retry mechanism in case of failure

4.5 Compliance Requirements

- Use AWS best practices
- Follow basic email sending policies

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5. Assumptions and Dependencies

5.1 Assumptions

- User has basic AWS knowledge
- Internet connectivity is available
- Required permissions are granted

5.2 Dependencies

- AWS services (SES, SNS, WorkMail)
- Grafana installation
- EC2 instance availability

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6. Risk Analysis

6.1 Risk Identification

Risk	Description
SES Sandbox	Limited email sending
SMTP Block	Ports may be blocked
Misconfiguration	Setup errors
Email Delay	Delivery delays

6.2 Risk Mitigation

Risk	Mitigation
SES Sandbox	Verify emails/domains
SMTP Block	Use alternate ports
Misconfiguration	Validate configs
Email Delay	Monitor logs

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7. Glossary

Term	Meaning
Alert	Notification triggered by condition
SMTP	Email sending protocol
SNS	Notification service

SES	Email sending service
WorkMail	Managed email system

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8. Appendices

- Architecture Diagram
- Configuration Screenshots
- Alert Email Screenshots

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9. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____